

AP Precalculus: 2026–2027

Unit 1: Polynomial and Rational Functions Lesson 1.12: Transformations of Functions

Primary Objective: Students will identify, describe, and construct transformations of functions using additive transformations (vertical and horizontal translations) and multiplicative transformations (vertical and horizontal dilations, including reflections), and will determine the domain and range of a transformed function from the parent’s domain and range.

Skills: MP 1.C, MP 3.A

Priority Ideas & Skills

AP Skills

Key Understandings

MP 1.C — Construct new functions as transformations of known functions.

- $g(x) = f(x) + k$: vertical translation by k units. (EK 1.12.A.1)
- $g(x) = f(x + h)$: horizontal translation by $-h$ units. (EK 1.12.A.2)
- $g(x) = af(x)$: vertical dilation by $|a|$; $a < 0$ also reflects over x -axis. (EK 1.12.A.3)
- $g(x) = f(bx)$: horizontal dilation by $1/|b|$; $b < 0$ also reflects over y -axis. (EK 1.12.A.4)

MP 3.A — Describe characteristics of transformed functions.

- Transformations can be combined: $g(x) = af(x + h) + k$. (EK 1.12.A.5)
- The domain and range of the transformed function may differ from the parent. (EK 1.12.A.6)

Vocabulary, Concepts, & Theorems

parent function

The original, un-transformed function from which transformed versions are built.

transformation

An operation that shifts, scales, or reflects the graph of a function.

vertical translation

$g(x) = f(x) + k$: shifts the graph up or down by $|k|$ units.

horizontal translation

$g(x) = f(x + h)$: shifts the graph left (if $h > 0$) or right (if $h < 0$) by $|h|$ units.

vertical dilation

$g(x) = af(x)$: scales outputs by $|a|$; $a < 0$ reflects over x -axis.

horizontal dilation

$g(x) = f(bx)$: scales inputs by $1/|b|$; $b < 0$ reflects over y -axis.

reflection

A special dilation with $|a| = 1$ or $|b| = 1$ and negative sign; flips the graph.

Activate Prior Knowledge & Spiral Review

Spiral review (warm-up) covers: evaluating $f(x)$, $f(-2)$, and $f(x + 1)$ from a formula (Lesson 1.1 function notation); reading domain and range from a table (Lesson 1.1); interpreting end behavior of a polynomial graph (Lessons 1.6–1.7, prerequisite for understanding how dilations affect overall shape).

Hook

“One function — four disguises. Can you un-transform it back?”

Display the table of $f(x) = x^2$ alongside tables for $f(x) + 3$, $f(x - 2)$, and $2f(x)$ without labeling them. Ask: “What did I *do* to f ? How do the outputs change? What stays the same?” Let students discuss for 60 seconds before revealing the rules. Emphasize the goal: given a rule, predict the table; given two tables, reverse-engineer the rule.

Lesson

Part 1 (12 min): Four Transformation Types.

Walk through each type in the guided notes with the $f(x) = x^2$ example:

1. **Vertical shift** $g(x) = f(x) + k$: output changes, input doesn't. Domain unchanged; range shifts.
2. **Horizontal shift** $g(x) = f(x + h)$: input changes, output values unchanged but at different x . Stress the direction reversal: $f(x + 3)$ shifts *left*.
3. **Vertical dilation** $g(x) = af(x)$: outputs scaled by a . Negative a flips.
4. **Horizontal dilation** $g(x) = f(bx)$: inputs scaled, graph compressed ($|b| > 1$) or stretched.

Part 2 (10 min): Combined Transformations.

Model $g(x) = -2f(x + 1) - 3$ step by step:

1. Replace x with $x + 1$ (shift left 1).
2. Multiply by -2 (stretch by 2, reflect over x -axis).
3. Subtract 3 (shift down 3).

Use a table-of-values approach to show how each step changes the outputs numerically.

Part 3 (5 min): Domain and Range.

For f with domain $[-2, 4]$ and range $[0, 9]$, work out domain/range of $g(x) = -2f(x + 1) - 3$: domain shifts left 1 $\rightarrow [-3, 3]$; range: $[0, 9] \times (-2) = [-18, 0]$, then $-3 \rightarrow [-21, -3]$. Emphasize that order matters when the factor is negative.

Active Monitoring

- **Key misconception to catch:** Students often shift $f(x + 3)$ *right* instead of left. Cold-call: “If $g(x) = f(x + 3)$, at what x -value does g equal $f(0)$? Why?” (Answer: $x = -3$, so the graph moved left.)
- Circulate during the notes table-fill: check that students compute $g(x) = 2f(x)$ by doubling the f column, not the x column.
- Watch for students who apply transformations in the wrong order for combined cases.
- After Part 3: ask a student to explain why domain shifts left when $h > 0$ in $f(x + h)$.

Group Work & Differentiation

Tier R — Remediate

- Pre-factored transformation tables provided.
- Match the completed table to the rule (multiple choice).
- Focus: reading how outputs change and connecting to the rule label.

Tier A — Apply

- Build the table for $g(x) = 2f(x - 1) + 3$ step by step.
- State domain and range of g .
- All computation shown in a scaffolded multi-column table.

Tier E — Extend

- Given tables for f and h , reverse-engineer the rule for h in terms of f .
- Identify a and k ; describe all transformations in words.

Individual Work & Assessment

Exit ticket (3 questions, 1 page):

1. Identify and describe the transformations in $g(x) = f(x + 3) - 2$ (tests horizontal shift direction, vertical shift).
2. Complete the table for $g(x) = -f(x)$ given f values (tests vertical reflection).
3. State the domain of $g(x) = f(x - 4)$ given domain of f is $[-2, 6]$ (tests domain shift).

Collect exit tickets; sort by correctness on Q1 (horizontal shift direction) to identify students needing re-teaching before Lesson 1.13.

Reinforcement & Extension

Homework: 5 problems covering all four transformation types (identification and description), completing transformation tables, domain/range under combined transformations, describing $g(x) = -3f(x + 2) + 1$ step by step, and an AP-style MC on range under $g(x) = -2f(x) + 1$.

Reflection: Students explain in their own words why horizontal transformations act opposite to the sign of h (builds conceptual understanding for Lesson 1.13).

Preview of Lesson 1.13: Building and selecting models for data — students will need to recognize parent function shapes and use transformation parameters to fit a function to a context. Transformations are the core tool for that lesson.